



Changing the subject with simple formula

Task A: Identifying the subject of the formula

1) Identify the **subject** in each of these formula/equations.

a) $C = \pi d$

b) $y = 7x - 5$

c) $\frac{C}{\pi} = d$

d) $A = \frac{1}{2}bh$

e) $h = \frac{A}{b}$

f) $\frac{F}{m} = a$

g) $d = s \times t$

h) $P = 2(b + h)$

2) Aisha is solving the equation $85 - 3x = 17$

The equation can be rearranged in these forms:

$$85 - 3x = 17$$

$$85 - 17 = 3x$$

$$85 = 17 + 3x$$

$$85 = 3x + 17$$

The form $85 - 17 = 3x$ is the best rearrangement because $3x$ is the **subject**.



You might have written:

*Whilst $85 - 17 = 3x$ is the best form from which to solve we can not call $3x$ the **subject** because the coefficient is 3 not 1*



Task B: Changing the subject of simple formula

1) In each equation/formula change the **subject** as instructed.

a) Make x the **subject**.

$$x - 2.7 = 8.1$$

$$x - 2.7 + 2.7 = 8.1 + 2.7$$

$$x = 8.1 + 2.7$$

$$x = 10.8$$

b) Make l the **subject**.

$$P = 3l$$

$$P \div 3 = 3 \times l \div 3$$

$$\frac{P}{3} = l$$

c) Make y the **subject**.

$$17 - y = 4$$

$$17 - y + y = 4 + y$$

$$17 = 4 + y$$

$$17 - 4 = 4 + y - 4$$

$$17 - 4 = y$$

$$13 = y$$

d) Make w the **subject**.

$$A = l \times w$$

$$A \div l = l \times w \div l$$

$$\frac{A}{l} = w$$

e) Make u the **subject**.

$$v = u + at$$

$$v - at = u + at - at$$

$$v - at = u$$

f) Make r the **subject**.

$$C = 2\pi r$$

$$C \div 2\pi = 2\pi r \div 2\pi$$

$$\frac{C}{2\pi} = r$$

2) Rearrange this formula as instructed.

$$F - E + V = 2$$

a) Make F the **subject**.

$$F - E + V = 2$$

$$F - E + V + E - V = 2 + E - V$$

$$F = 2 + E - V$$

b) Make V the **subject**.

$$F - E + V = 2$$

$$F - E + V + E - F = 2 + E - F$$

$$V = 2 + E - F$$

c) Make E the **subject**.

$$F - E + V = 2$$

$$F - E + V + E = 2 + E$$

$$F + V = 2 + E$$

$$F + V - 2 = 2 + E - 2$$

$$F + V - 2 = E$$

